

Based on the provided sources, here are the solutions to the questions presented in the "Mid Sem 2024 - 25.pdf" examination paper:

Section A

Q1. Discuss the operational challenges associated with thick seam mining. Enumerate the typical ground control requirements for mining thick seams.

- **Operational Challenges:**
- **Reduced Pillar Strength:** As the extraction height increases for a given pillar width, the width-to-height (W/H) ratio is lowered. This inherently reduces the structural stiffness and overall strength of the coal pillar 1.
- **Pillar and Face Spalling:** The reduced stiffness of the thicker coal structure lowers its resistance to deformation under load, making the face highly susceptible to face spalling and sloughing 2.
- **En-masse Caving & Severe Weighting:** Thicker mining increases the height of the Natural Pressure Arch (NPA) into the goaf. Consequently, a much larger volume of rock participates in the caving process, causing high-volume "en-masse" caving. This violent caving throws a massive, strong dynamic weight onto the working area and face supports 2, 3.
- **Ground Control Requirements:**
- Must deploy high-capacity powered roof supports that can withstand intense periodic weighting and dynamic loading without irreparable damage 2, 4.
- Supports must utilize face sprags/flippers to actively hold the coal face and prevent severe sloughing or cavity formation 2, 5.
- Pillars must be designed with larger dimensions to maintain a safe SF (Stability Factor) compensating for the loss of W/H ratio 1.
- *Required Schematics:* (1) A sketch illustrating the "Effect of mining height on pillar strength" showing the pillar strength decreasing as height h_1 to h_3 increases 6. (2) A sketch of the "Pressure Arch Formation Mechanism" showing the stress relief zone above the coal and abutment stress loading the surrounding pillars 3.

Q2.a. What are the different zones that form above the coal seam with the progressive mining in a typical longwall working? How do the dynamics of caving alter with the variation of bulking factor, strata characteristics, and mining height?

- **Zones Formed:** The overburden is disturbed in order of severity from the seam to the surface, creating four zones 7-9:
- *Caved Zone:* Irregularly broken strata that fall directly onto the floor (thickness ranges from 2 to 8 times the mining height).
- *Fractured Zone:* Broken into blocks by vertical/horizontal cracks but partially maintain contact (thickness ranges from 28 to 52 times the mining height).
- *Continuous Deformation Zone:* Strata deform gently behaving like an intact medium without major cracks cutting through.
- *Soil Zone:* The surface layer where permanent cracks and steps may appear.
- **Dynamics of Caving:**

- **Bulking Factor:** As roof rocks cave, their volume increases (bulks). A higher bulking factor means the caved rocks fill the goaf void faster. This reduces the caving height required to support the overlying main roof, minimizing the severity of the main fall 10.
- **Strata Characteristics:** Weak, highly laminated strata cave immediately after the shield advances. In contrast, strong/massive strata (like thick sandstones) will overhang for large spans into the goaf, accumulating massive strain energy until they snap, causing severe cyclic periodic weighting 11, 12.
- **Mining Height:** As the mining height increases, the caved and fractured zones extend much higher into the roof. A greater volume of rock participates in the caving, which triggers high-volume/en-masse caving and throws heavier abutment stress to the face 2, 7.

Q2.b. Discuss the supporting cycle and loading characteristics of the powered roof support deployed in the longwall operation.

1. **Supporting Cycle & Loading Characteristics:** A mining cycle varies from cycle to cycle due to roof interaction, but typically consists of the following distinctive periods 13, 14:
2. **Setting Period (o-s):** The shield advances and sets against the roof, rapidly achieving an initial setting pressure (\$P_s\$).
3. **Rapid Increase Period (s-a):** The support reacts to the descending roof, and resistance increases rapidly until reaching temporary relative equilibrium.
4. **Relatively Stable Period (a-b):** The longest period of the cycle. Occurs while the shearer is operating far away, resulting in a slow, stable increase in roof loading.
5. **Cutting Influenced Period (b-c):** As the shearer cuts past the support, exposing an additional web width of unsupported roof, the loading increases significantly.
6. **Neighboring Shield Movement (c-d):** When the adjacent shield disengages to advance, its load transfers suddenly to the support, causing a sharp spike until it hits the Final pressure (\$P_f\$).
7. **Release (d-e):** The shield is lowered to advance, dropping the resistance to zero.
8. **Required Schematic:** A line graph showing "Support Resistance (y-axis) vs. Time (x-axis)". The curve should start at \$P_s\$, slope steeply to '\$a\$', flatten out from '\$a\$' to '\$b\$', steepen from '\$b\$' to '\$c\$', spike sharply to maximum final pressure '\$d\$', and drop vertically to zero at '\$e\$' 13, 14.

Q3. With neat sketches, describe the abutment stress distribution about the longwall panel.

- **Distribution Description:**
- **Front Abutment Stress:** Concentrated in the solid coal directly ahead of the advancing face. It peaks slightly ahead of the face and gradually decays to the in-situ stress level outby. The magnitude is highest in the central region of the face length and causes peak-stress conditions just before weighting 15-17.
- **Side Abutment Stress:** Concentrated along both sides of the panel on the chain pillars and unmined ribs in the goaf. It is highest at the ribs and drops exponentially moving away from the active panel 17, 18.

- **Corner Peak Stress:** The face corners (T-junctions) experience extreme stress concentration due to the superimposition (interaction) of the front and side abutment stresses 17.
- **Goaf Stress:** Immediately behind the shields, the stress is zero due to lack of compaction. As the gob material touches the overlying strata and compacts, stress re-establishes and reaches maximum cover pressure further back in the waste area 17, 19.
- **Required Schematic:** A 3D isometric sketch of a longwall panel showing the "Typical stress redistribution" with peaks at the face corners (T-junctions), a wave representing front abutment ahead of the face, side abutment waves over the pillars, and the stress recovering in the goaf behind the face 17.

Q4.a. What is 'Roof Cavity'? Discuss the mechanism that leads to the formation of the roof cavity.

- **Roof Cavity:** A roof cavity is a localized void formed at the longwall face when the immediate roof fractures and falls out prematurely before the powered roof support can advance to hold it 5.
- **Mechanism:** Cavities typically form in weak or friable immediate roofs in the unsupported span between the canopy tip and the solid coal face. The excavation induces horizontal tensile stresses in this exposed span. Because rock is extremely weak in tension, horizontal cracks initiate and cause the strata to fall. The likelihood of cavity formation drastically increases if the powered supports (e.g., 4-leg shields) fail to apply an adequate *active horizontal compressive force* to the roof during setting, or if the canopy tip-to-faceline distance is excessively wide, leaving the weak beam unsupported 5, 20-23.
- **Required Schematic:** A cross-section of a longwall face showing a shield canopy, the coal face, and a highly fractured hollow void (cavity) in the immediate roof spanning between the canopy tip and the faceline 5.

Q4.b. Enumerate the mechanism of face damage and various forms of damage parameters.

- **Mechanism:** Before excavation, the coal mass behaves elastically. Extracting coal redistributes stresses, triggering immense stress concentrations (abutment stresses) around the working face. Depending on the intensity of this interaction between the maximum abutment and minimum confining stresses, and the associated strain rate, the coal face degrades into various states of irreversible plastic failure or elastic yielding 24-26.
- **Damage Parameters/Forms:**
- **Highly Damaged State/Zone (HDS/HDZ):** An irreversible plastic region right at the free face. It represents coal mass that completely detaches or violently slabs off (spalls) into the walkway 25.
- **Excavation Damaged State/Zone (EDS/EDZ):** The fractured/failed rock mass extending deeper into the opening behind the HDZ. The rock's properties are irreversibly altered by the maximum governing stress 26.

- *Excavation Influence State/Zone (EIS/EIZ)*: The zone immediately beyond the EDZ deeper in the solid coal. Changes here are insignificant and reversible (strata retains elastic behavior) 26.
- *Required Schematic*: The "Axial Strain vs. Volumetric Strain" graph and the top-down section showing the coal face divided into the Intact Region, Region in EDS (Plastic Region), and Region in HDS (with spalling) 25.

Section B

Q5. Estimate the span of the main fall and answer related queries. Given Formula: $W_s = 324 \cdot F_L^{-0.43} \cdot D^{-1.38} \cdot \sqrt{\frac{\sigma_{cIR} \cdot T_{IR}}{\gamma_{IR}}} \cdot \sqrt{\frac{\sigma_{cMR} \cdot T_{MR}}{\gamma_{MR}}}$ **\$5.a. Calculation for Mine 'A':**

- **Given:** $F_L = 200\text{m}$, Depth $D = 350\text{m}$. The note dictates taking $\gamma = 0.023\text{ MPa/m}$ for unit weight.
- **Properties from Table 1 27:**
- Immediate Roof (IR): $\sigma_{cIR} = 12.25\text{ MPa}$, $T_{IR} = 1.10\text{ MPa}$.
- Main Roof (MR): $\sigma_{cMR} = 12.89\text{ MPa}$, $T_{MR} = 1.13\text{ MPa}$.
- **Step 1: Compute Immediate Roof component:** $\sqrt{\frac{12.25 \cdot 1.10}{0.023}} = \sqrt{\frac{13.475}{0.023}} = \sqrt{585.87} = 24.205$
- **Step 2: Compute Main Roof component:** $\sqrt{\frac{12.89 \cdot 1.13}{0.023}} = \sqrt{\frac{14.5657}{0.023}} = \sqrt{633.29} = 25.165$
- **Step 3: Compute F_L and D components:** $200^{-0.43} = 0.1037$, $350^{-1.38} = 0.0003253$
- **Step 4: Final calculation:** $W_s = 324 \cdot 0.1037 \cdot 0.0003253 \cdot 24.205 \cdot 25.165 \approx 6.65\text{ meters}$. **Answer:** The estimated span of the main fall is **\$6.65 m**.

5.b. Calculation for altered parameters:

- **Given:** New $F_L = 250\text{m}$, New $D = 450\text{m}$. Strata characteristics remain the same.
- **Calculation:** We can solve via proportionality: $W_{s2} = W_{s1} \cdot \left(\frac{250}{200}\right)^{-0.43} \cdot \left(\frac{450}{350}\right)^{-1.38}$
 $W_{s2} = 6.65 \cdot (1.25)^{-0.43} \cdot (1.2857)^{-1.38} = 6.65 \cdot 0.908 \cdot 0.707 \approx 4.27\text{ meters}$. **Answer:** The span of the main fall for the new panel is **\$4.27 m**.

5.c. Why are coal strength and mining height not considered in the relation?

- **Answer:** The main fall span specifically calculates the structural failure limit of the massive, strong **main roof** plate overhanging the goaf. While coal strength and mining height deeply influence the caving of the *immediate roof* (filling the void) and the resulting spalling/convergence at the face itself, the ultimate snapping point of the main roof plate depends entirely on its own span, thickness, tensile/compressive strength, and the in-situ overburden pressure (depth) driving it down 11, 28, 29.

Q6. Estimate the support capacity for Mine 'B'. Given Formulas: Mean Load Density: $C_m = \left(\frac{A}{P} + 9.6h_e\right) + \left(\frac{k \cdot I_{\max}}{k_1 + 1.5} - 23\right)$ 30, 31 Support Capacity (T): $P_r = P_c \cdot W_c \cdot (L_c + W_d + 0.3)$ 32, 33 Rated Support Resistance $P_r = P \cdot 1.3$ 32

6.a. Calculation for Initial Parameters:

- **Given parameters:** Depth \$= 150\text{m}\$, Face length \$= 180\text{m}\$, \$h_e = 3.5\text{m}\$, \$W_c = 1.75\text{m}\$, \$L_c = 4\text{m}\$, \$W_d = 0.8\text{m}\$. For Indian geo-mining conditions, maximum acceptable convergence \$C_m = 60\text{ mm/m}\$ 34. Constant \$k = 0.025\$ 35.
- **Determine variables from Table 2 36:**
- Caving zone height is generally assumed to be 15 times extraction height \$= 15 \times 3.5 = 52.5\text{m}\$ 37.
- Looking at Table 2, all beds up to Bed 14 (depth \$48.35\text{m}\$ above seam) fall inside the \$52.5\text{m}\$ caving zone limit.
- The highest Caving Index (\$I_{\max}\$) belongs to Bed 14 with **\$I_{\max} = 6197\$**.
- \$I_{\max} = 6197\$ places the roof in **Category III** (\$5000 \leq I < 10000\$) 38. For Category III, empirical constant **\$A = 1700\$** 30.
- Thickness of caving beds (\$t\$) between the strong bed (Bed 14) and coal seam is the sum of thicknesses of Beds 1 through 13. By looking at depth above coal seam for the bottom of Bed 14, \$t = 40.76\text{m}\$ 36.
- Ratio \$r = t / h_e = 40.76 / 3.5 = 11.64\$. Since \$r > 8\$, the factor **\$k_1 = 10\$** 30.
- **Step 1: Calculate Mean Load Density (\$P\$):**

$$P = \left(\frac{1700}{t} + 9.6 \right) + \left(\frac{0.025 \times 6197}{10 + 1.5} - 23 \right)$$

$$P = \frac{1700}{40.76} + 33.6 + \frac{154.925}{11.5} - 23$$

$$P = \frac{1700}{40.76} + 33.6 + 13.47 - 23$$

$$P = \frac{1700}{40.76} + 24.07$$

$$35.93 = \frac{1700}{P} \implies P = \frac{1700}{35.93} = 47.31 \text{ t/m}^2$$
- **Step 2: Calculate Rated Support Resistance (\$P_r\$):**

$$P_r = 1.3 \times 47.31 = 61.5 \text{ t/m}^2$$
- **Step 3: Calculate Support Capacity:**

$$\text{Capacity} = 61.5 \times 1.75 \times (4 + 0.8 + 0.3) = 61.5 \times 1.75 \times 5.1 \approx 548.9 \text{ tonnes}$$

6.b. Altering face height to \$h_e = 4.5\text{m}\$:

- New \$r = t / h_e = 40.76 / 4.5 = 9.05\$. Since \$r > 8\$, \$k_1 = 10\$ still applies. \$A = 1700\$.
- **Calculate \$P\$:**

$$P = \left(\frac{1700}{t} + 9.6 \right) + 13.47 - 23$$

$$P = \frac{1700}{40.76} + 43.2 - 9.53$$

$$P = \frac{1700}{40.76} + 33.67$$

$$26.33 = \frac{1700}{P} \implies P = 64.56 \text{ t/m}^2$$
- **Calculate \$P_r\$ and Capacity:**

$$P_r = 1.3 \times 64.56 = 83.93 \text{ t/m}^2$$

$$\text{Capacity} = 83.93 \times 1.75 \times 5.1 \approx 749.1 \text{ tonnes}$$

6.c. If \$I_{\max} = 4915\$ and \$t = 31\text{m}\$ (with \$h_e = 3.5\text{m}\$):

- \$I_{\max} = 4915\$ places the roof in **Category II** (\$2000 \leq I < 5000\$) 38. For Category II, **\$A = 1440\$** 30.
- New \$r = 31 / 3.5 = 8.85\$. Since \$r > 8\$, **\$k_1 = 10\$**.
- **Calculate \$P\$:**

$$P = \left(\frac{1440}{t} + 9.6 \right) + \left(\frac{0.025 \times 4915}{11.5} - 23 \right)$$

$$P = \frac{1440}{31} + 33.6 + (10.68 - 23)$$

$$P = \frac{1440}{31} + 33.6 - 12.32$$

$$P = \frac{1440}{31} + 21.28$$

$$38.72 = \frac{1440}{P} \implies P = \frac{1440}{38.72} = 37.19 \text{ t/m}^2$$
- **Calculate \$P_r\$ and Capacity:**

$$P_r = 1.3 \times 37.19 = 48.35 \text{ t/m}^2$$

$$\text{Capacity} = 48.35 \times 1.75 \times 5.1 \approx 431.5 \text{ tonnes}$$

